

instrument based on the virtual boundary to avoid the detected object.

21 Claims, 7 Drawing Sheets

Related U.S. Application Data

continuation of application No. 16/685,442, filed on Nov. 15, 2019, now Pat. No. 11,464,579, which is a continuation of application No. 15/416,717, filed on Jan. 26, 2017, now Pat. No. 10,512,509, which is a division of application No. 14/205,702, filed on Mar. 12, 2014, now Pat. No. 9,603,665.

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See application file for complete search history.

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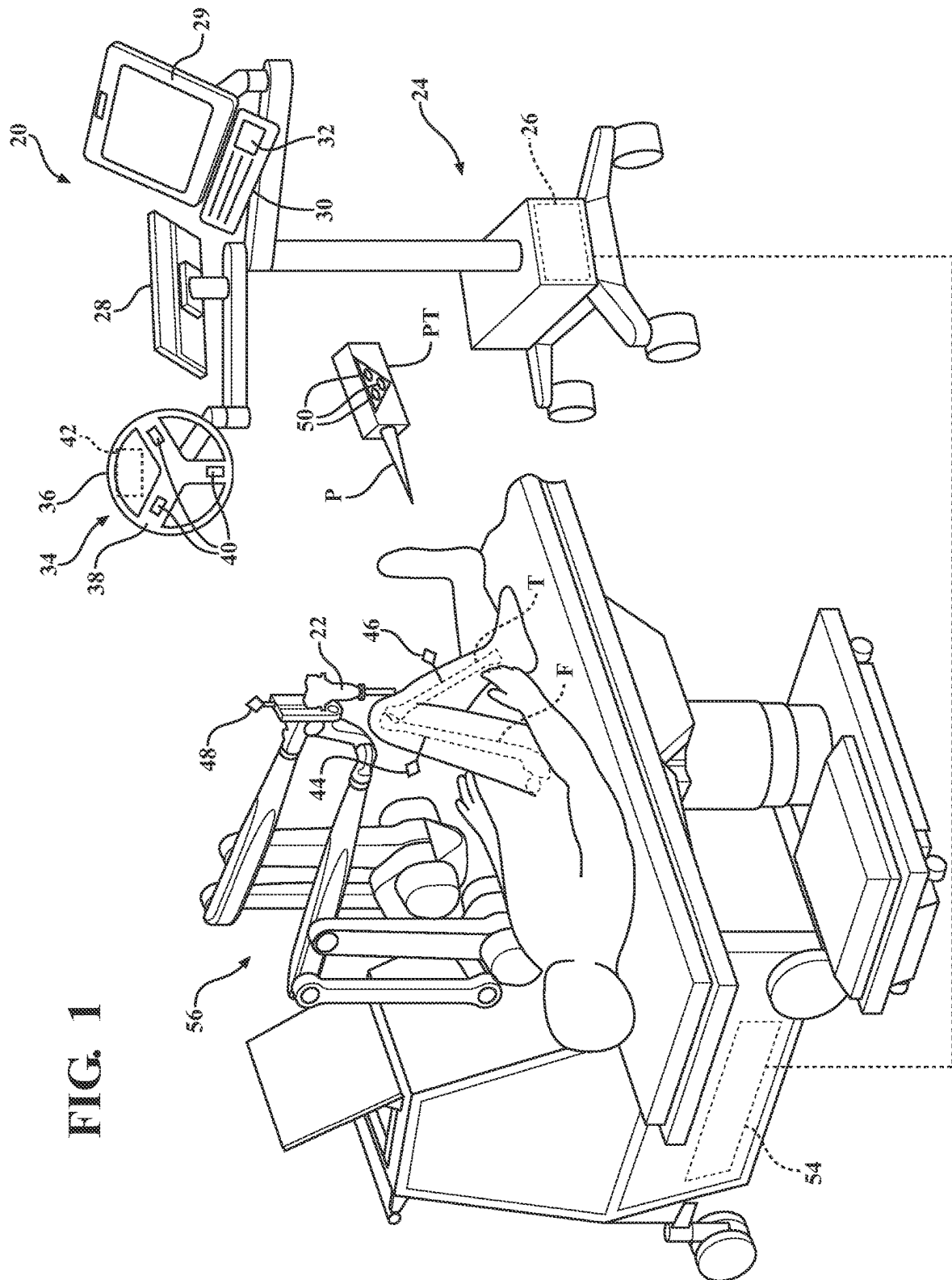
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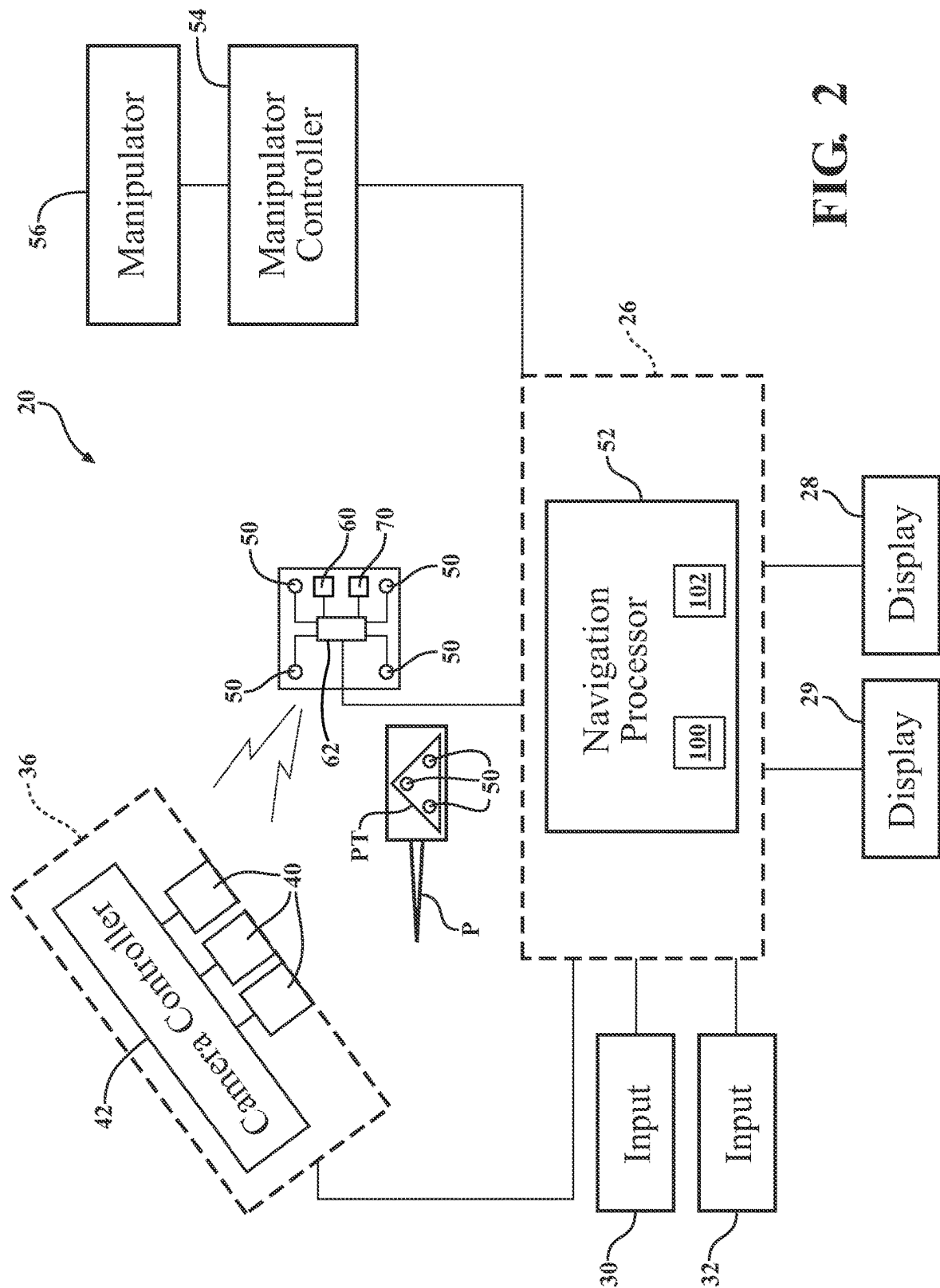


FIG. 2

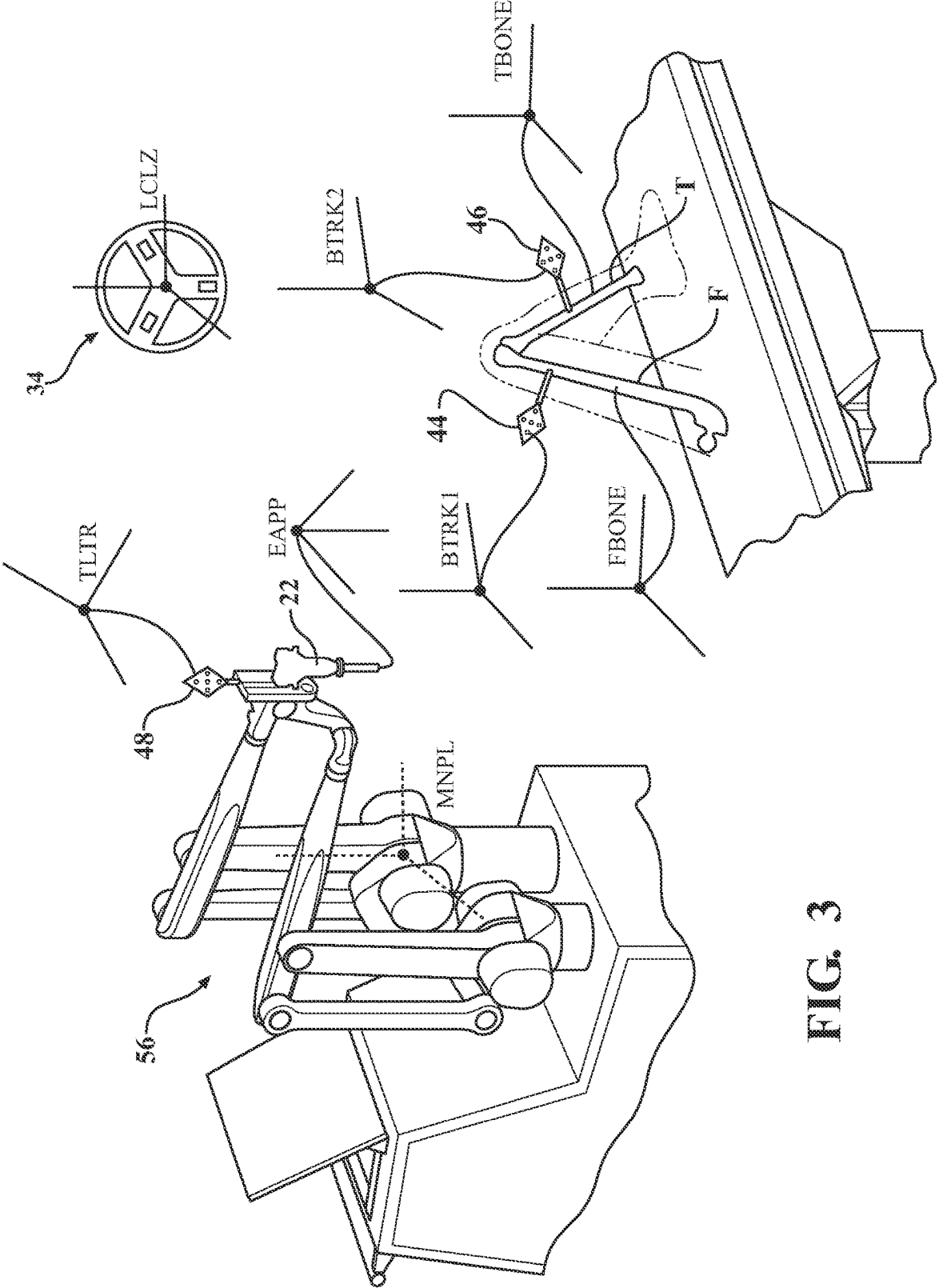


FIG. 3

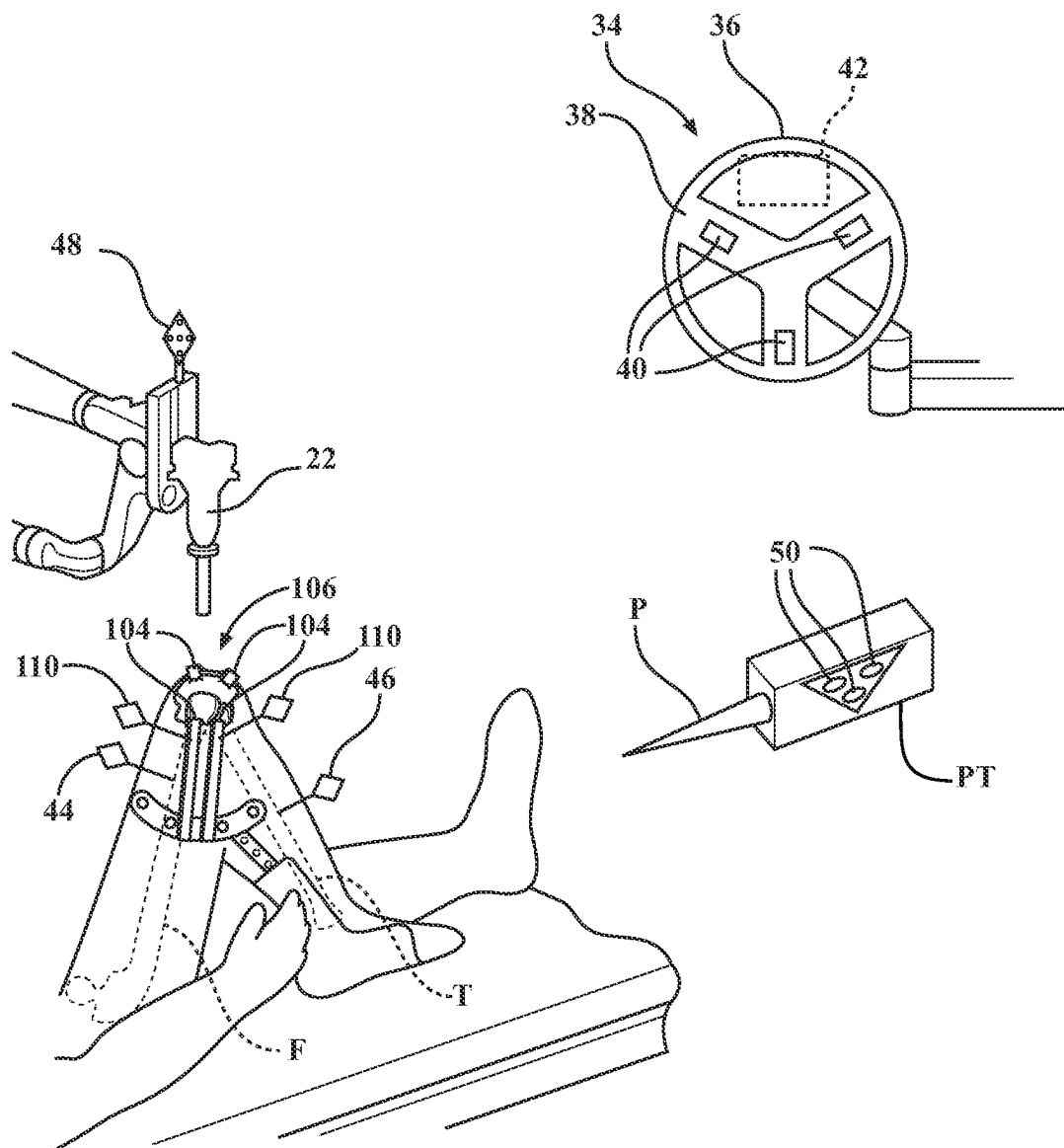


FIG. 4

FIG. 5

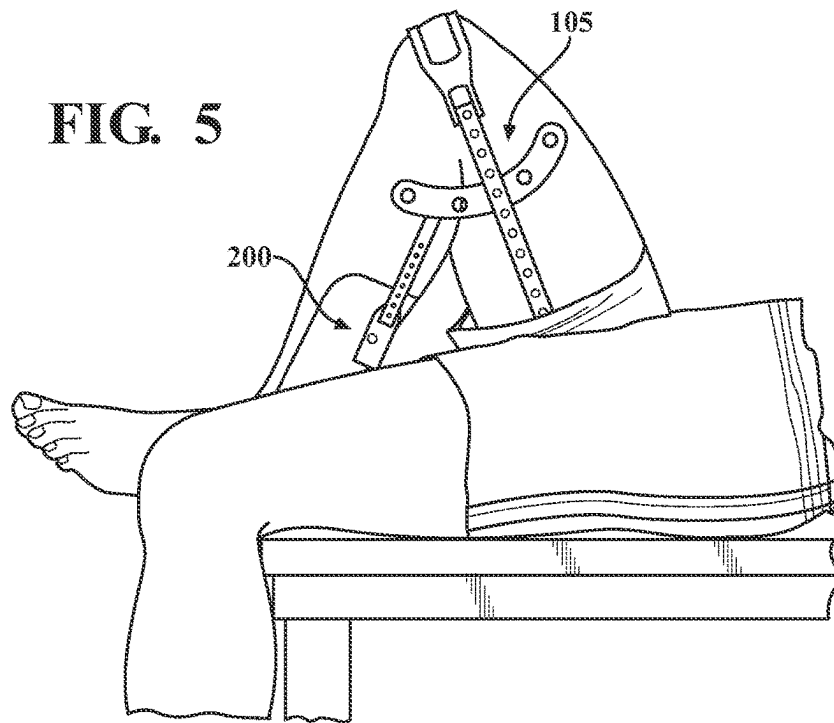


FIG. 6

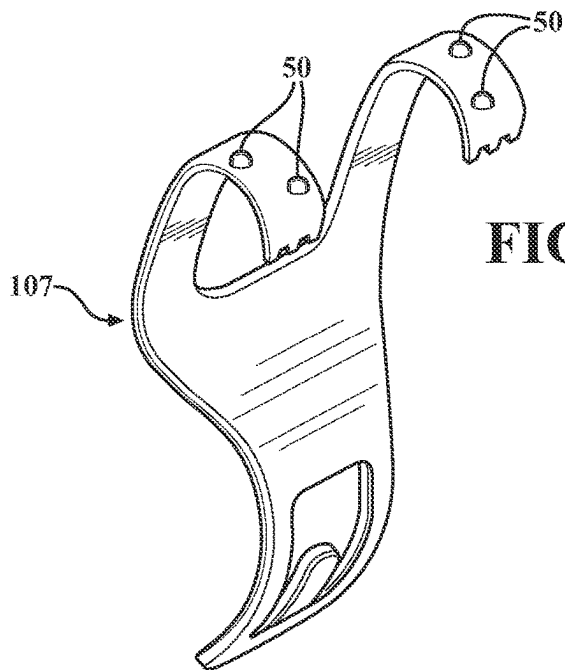


FIG. 7

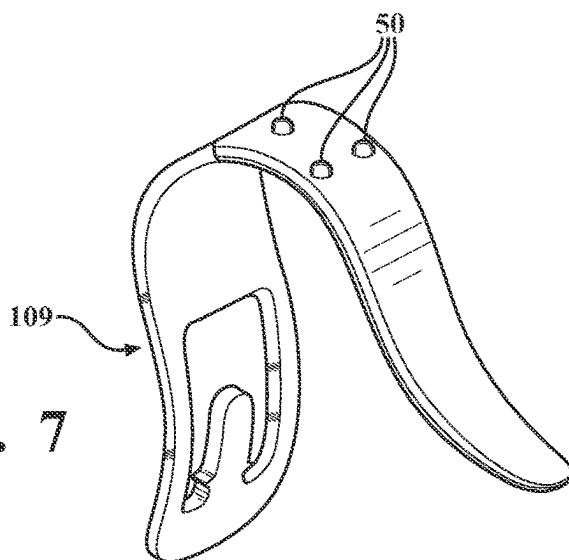
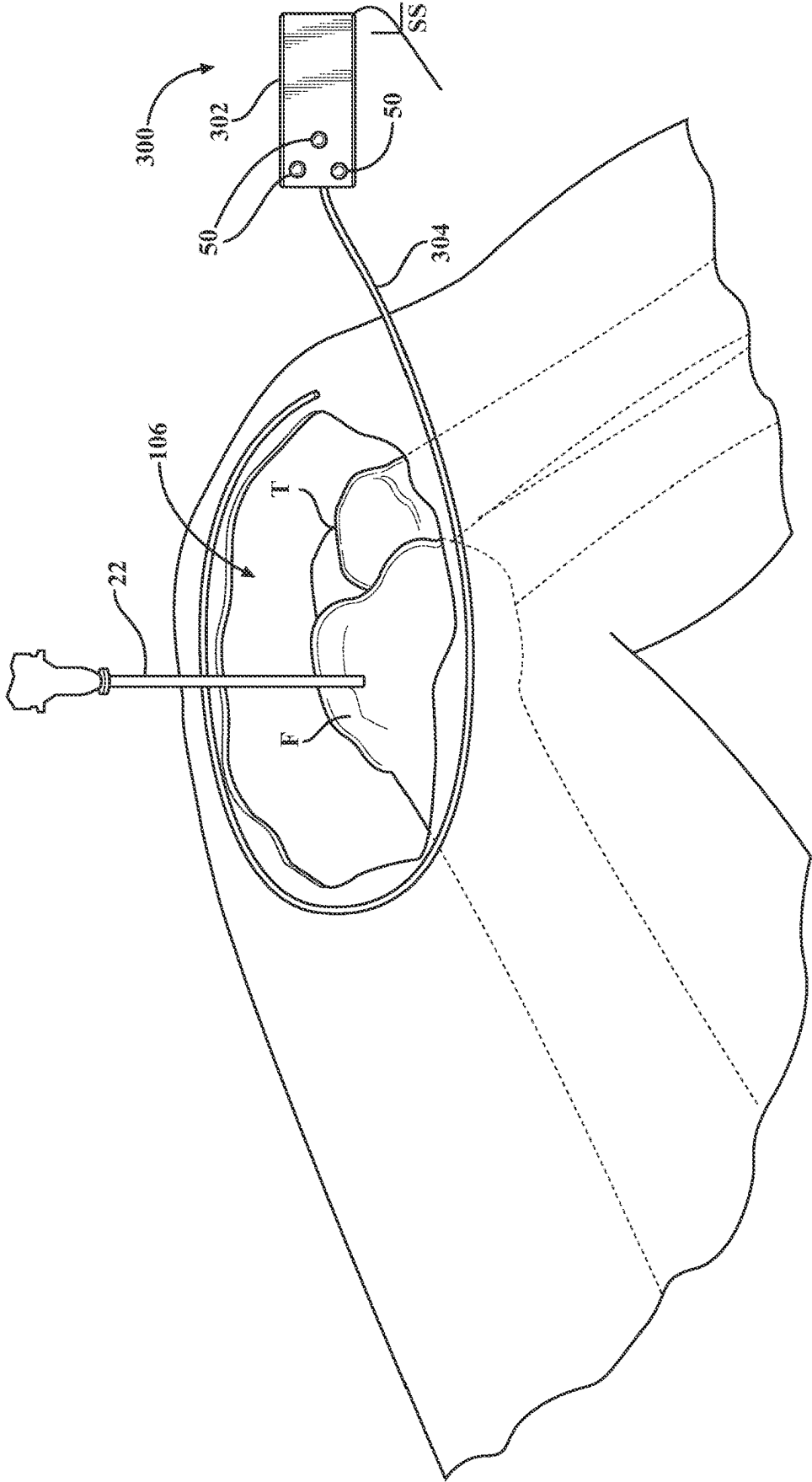


FIG. 8



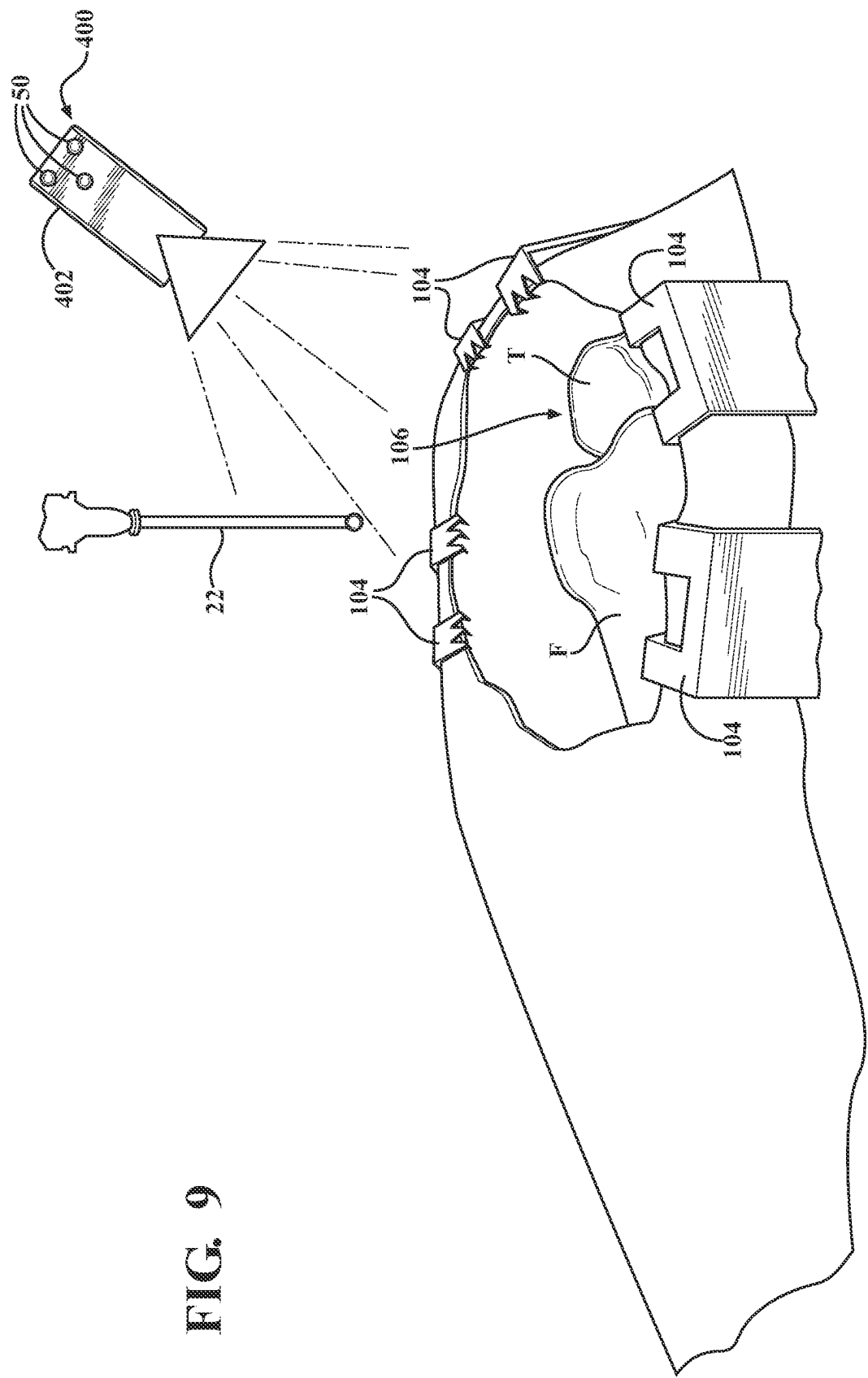


FIG. 9

SYSTEMS AND METHODS FOR ESTABLISHING VIRTUAL CONSTRAINT BOUNDARIES

RELATED APPLICATIONS

The subject application is a continuation of U.S. patent application Ser. No. 17/891,288, filed Aug. 19, 2022, which is a continuation of U.S. patent application Ser. No. 16/685,442, filed Nov. 15, 2019, now U.S. Pat. No. 11,464,579, which is a continuation of U.S. patent application Ser. No. 15/416,717, filed Jan. 26, 2017, now U.S. Pat. No. 10,512,509, which is a division of U.S. patent application Ser. No. 14/205,702 filed on Mar. 12, 2014, now U.S. Pat. No. 9,603,665, which claims the benefit of U.S. Provisional Patent App. No. 61/780,148, filed on Mar. 13, 2013, the contents of each of the above-referenced applications being hereby incorporated by reference in their entirety.

TECHNICAL FIELD

The present disclosure relates generally to systems and methods for establishing and tracking virtual boundaries.

BACKGROUND

In robotic surgery virtual boundaries are created using computer aided design software to delineate areas in which an end effector of a robotic system can maneuver from areas in which the end effector is restricted. For instance, in orthopedic surgery a virtual cutting boundary may be created to delineate sections of bone to be removed by the end effector during the surgery from sections of bone that are to remain after the surgery.

A navigation system tracks movement of the end effector with respect to the virtual cutting boundary to determine a position and/or orientation of the end effector relative to the virtual cutting boundary. The robotic system cooperates with the navigation system to guide movement of the end effector so that the end effector does not move beyond the virtual cutting boundary.

Typically, virtual cutting boundaries are created prior to surgery. Virtual cutting boundaries are often created in a model of a patient's bone and fixed with respect to the bone so that when the model is loaded into the navigation system, the navigation system can track movement of the virtual cutting boundary by tracking movement of the bone.

Virtual boundaries may define other anatomical features to be avoided by the end effector during surgery. Such features include nerves or other types of tissue to be protected from contact with the end effector. Virtual boundaries are also used to provide virtual pathways that direct the end effector toward the anatomy being treated. These examples of virtual boundaries are often fixed in relationship to the anatomy being treated so that all of the boundaries are tracked together as the anatomy moves. However, some anatomical features or other objects in the operating room may move relative to the anatomy being treated. For instance, retractors used to provide an opening in tissue for the end effector may move relative to the anatomy being treated. If not accurately tracked using an appropriate dynamic virtual constraint boundary, the end effector may inadvertently strike the retractors. As a result, the end effector may be damaged or become inoperative and the retractor may become dislodged from its position.

Other typically untracked objects may also be in proximity to the end effector that should be avoided by the end

effector, yet move relative to the anatomy being treated. Therefore, there is a need in the art for systems and methods for creating dynamic virtual boundaries for such objects.

SUMMARY

According to a first aspect, a surgical system is provided comprising: a robotic manipulator configured to support and move a cutting instrument; a machine vision system including a vision camera; and a control system coupled to the robotic manipulator and the machine vision system, wherein the control system is configured to: control the robotic manipulator to move the cutting instrument along a path to cut away material from a bone at a surgical site; detect, with the machine vision system, an object at, or in proximity to, the surgical site; associate a virtual boundary with the object detected by the machine vision system; adjust the path of the cutting instrument based on the virtual boundary; and control the robotic manipulator to move the cutting instrument along the adjusted path to enable the cutting instrument to avoid the object.

According to a second aspect, a method is provided of operating a surgical system, the surgical system including a robotic manipulator configured to support and move a cutting instrument, a machine vision system including a vision camera, and a control system coupled to the robotic manipulator and the machine vision system, the method comprising the control system: controlling the robotic manipulator for moving the cutting instrument along a path for cutting away material from a bone at a surgical site; detecting, with the machine vision system, an object at, or in proximity to, the surgical site; associating a virtual boundary with the object detected by the machine vision system; adjusting the path of the cutting instrument based on the virtual boundary; and controlling the robotic manipulator for moving the cutting instrument along the adjusted path and for enabling the cutting instrument to avoid the object.

According to a third aspect, a surgical system is provided comprising: a robotic manipulator configured to support and move a cutting instrument; a machine vision system including a vision camera; and a control system coupled to the robotic manipulator and the machine vision system, wherein the control system is configured to: control the robotic manipulator to move the cutting instrument to cut away material from a bone at a surgical site; detect, with the machine vision system, an object at, or in proximity to, the surgical site; associate a virtual boundary with the object detected by the machine vision system; and control the robotic manipulator to automatically adjust an orientation of the cutting instrument based on the virtual boundary to enable the cutting instrument to avoid the object.

According to a fourth aspect, a method is provided of operating a surgical system, the surgical system including a robotic manipulator configured to support and move a cutting instrument, a machine vision system including a vision camera, and a control system coupled to the robotic manipulator and the machine vision system, the method comprising the control system: controlling the robotic manipulator for moving the cutting instrument for cutting away material from a bone at a surgical site; detecting, with the machine vision system, an object at, or in proximity to, the surgical site; associating a virtual boundary with the object detected by the machine vision system; and controlling the robotic manipulator for automatically adjusting an

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orientation of the cutting instrument based on the virtual boundary to enable the cutting instrument to avoid the object.

BRIEF DESCRIPTION OF THE DRAWINGS

Advantages of the present invention will be readily appreciated as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings wherein:

FIG. 1 is a perspective view of a navigation system of the present invention being used in conjunction with a robotic system;

FIG. 2 is a schematic view of the navigation system;

FIG. 3 is schematic view of the coordinate systems used in the navigation system;

FIG. 4 is a perspective view of a tissue opening for accessing a knee joint by an end effector of the robotic system;

FIG. 5 is an elevational view of a leg holder and retractor assembly being used to maintain the tissue opening;

FIG. 6 is a top perspective view of a retractor;

FIG. 7 is a top perspective view of an alternative retractor;

FIG. 8 is a top perspective view of the tissue opening showing an end effector in the tissue opening and a flexible shape sensing device for tracking movement of the tissue opening; and

FIG. 9 is a top perspective view of the tissue opening showing an end effector in the tissue opening and a machine vision system for tracking movement of the tissue opening.

DETAILED DESCRIPTION

Referring to FIG. 1 a surgical navigation system 20 is illustrated. The system 20 is shown in a surgical setting such as an operating room of a medical facility. The navigation system 20 is set up to track movement of various objects in the operating room. Such objects include, for example, a surgical instrument 22, a femur F of a patient, and a tibia T of the patient. The navigation system 20 tracks these objects for purposes of displaying their relative positions and orientations to the surgeon and, in some cases, for purposes of controlling or constraining movement of the surgical instrument 22 relative to virtual cutting boundaries associated with the femur F and tibia T.

The surgical navigation system 20 includes a computer cart assembly 24 that houses a navigation computer 26. A navigation interface is in operative communication with the navigation computer 26. The navigation interface includes a first display 28 adapted to be situated outside of the sterile field and a second display 29 adapted to be situated inside the sterile field. The displays 28, 29 are adjustably mounted to the computer cart assembly 24. First and second input devices 30, 32 such as a keyboard and mouse can be used to input information into the navigation computer 26 or otherwise select/control certain aspects of the navigation computer 26. Other input devices are contemplated including a touch screen (not shown) or voice-activation.

A localizer 34 communicates with the navigation computer 26. In the embodiment shown, the localizer 34 is an optical localizer and includes a camera unit 36 (one example of a sensing device). The camera unit 36 has an outer casing 38 that houses one or more optical position sensors 40. In some embodiments at least two optical sensors 40 are employed, preferably three. The optical sensors 40 may be three separate charge-coupled devices (CCD). In one embodiment three, one-dimensional CCDs are employed. It

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should be appreciated that in other embodiments, separate camera units, each with a separate CCD, or two or more CCDs, could also be arranged around the operating room. The CCDs detect infrared (IR) signals.

Camera unit 36 is mounted on an adjustable arm to position the optical sensors 40 with a field of view of the below discussed trackers that, ideally, is free from obstructions. In some embodiments the camera unit 36 is adjustable in at least one degree of freedom by rotating about a rotational joint. In other embodiments, the camera unit 36 is adjustable about two or more degrees of freedom.

The camera unit 36 includes a camera controller 42 in communication with the optical sensors 40 to receive signals from the optical sensors 40. The camera controller 42 communicates with the navigation computer 26 through either a wired or wireless connection (not shown). One such connection may be an IEEE 1394 interface, which is a serial bus interface standard for high-speed communications and isochronous real-time data transfer. The connection could also use a company specific protocol. In other embodiments, the optical sensors 40 communicate directly with the navigation computer 26.

Position and orientation signals and/or data are transmitted to the navigation computer 26 for purposes of tracking objects. The computer cart assembly 24, display 28, and camera unit 36 may be like those described in U.S. Pat. No. 7,725,162 to Malackowski, et al. issued on May 25, 2010, entitled "Surgery System", hereby incorporated by reference.

The navigation computer 26 can be a personal computer or laptop computer. Navigation computer 26 has the display 28, central processing unit (CPU) and/or other processors, memory (not shown), and storage (not shown). The navigation computer 26 is loaded with software as described below. The software converts the signals received from the camera unit 36 into data representative of the position and orientation of the objects being tracked.

Navigation system 20 includes a plurality of tracking devices 44, 46, 48, also referred to herein as trackers. In the illustrated embodiment, one tracker 44 is firmly affixed to the femur F of the patient and another tracker 46 is firmly affixed to the tibia T of the patient. Trackers 44, 46 are firmly affixed to sections of bone. Trackers 44, 46 may be attached to the femur F and tibia T in the manner shown in U.S. Pat. No. 7,725,162, hereby incorporated by reference. Trackers 44, 46 could also be mounted like those shown in U.S. patent application Ser. No. 14/156,856, filed on Jan. 16, 2013, entitled, "Navigation Systems And Methods For Indicating And Reducing Line-of-sight Errors", hereby incorporated by reference herein. In additional embodiments, a tracker (not shown) is attached to the patella to track a position and orientation of the patella. In yet further embodiments, the trackers 44, 46 could be mounted to other tissue types or parts of the anatomy.

An instrument tracker 48 is firmly attached to the surgical instrument 22. The instrument tracker 48 may be integrated into the surgical instrument 22 during manufacture or may be separately mounted to the surgical instrument 22 in preparation for the surgical procedures. The working end of the surgical instrument 22, which is being tracked by virtue of the instrument tracker 48, may be a rotating bur, electrical ablation device, or the like.

The trackers 44, 46, 48 can be battery powered with an internal battery or may have leads to receive power through the navigation computer 26, which, like the camera unit 36, preferably receives external power.

In the embodiment shown, the surgical instrument **22** is attached to a surgical manipulator. Such an arrangement is shown in U.S. patent application Ser. No. 13/958,070, entitled, "Surgical Manipulator Capable of Controlling a Surgical Instrument in Multiple Modes", the disclosure of which is hereby incorporated by reference.

In other embodiments, the surgical instrument **22** may be manually positioned by only the hand of the user, without the aid of any cutting guide, jig, or other constraining mechanism such as a manipulator or robot. Such a surgical instrument is described in U.S. patent application Ser. No. 13/600,888, filed Aug. 31, 2012, entitled, "Surgical Instrument Including Housing, a Cutting Accessory that Extends from the Housing and Actuators that Establish the Position of the Cutting Accessory Relative to the Housing", hereby incorporated by reference.

The optical sensors **40** of the localizer **34** receive light signals from the trackers **44**, **46**, **48**. In the illustrated embodiment, the trackers **44**, **46**, **48** are active trackers. In this embodiment, each tracker **44**, **46**, **48** has at least three active tracking elements or markers for transmitting light signals to the optical sensors **40**. The active markers can be, for example, light emitting diodes or LEDs **50** transmitting light, such as infrared light. The optical sensors **40** preferably have sampling rates of 100 Hz or more, more preferably 300 Hz or more, and most preferably 500 Hz or more. In some embodiments, the optical sensors **40** have sampling rates of 8000 Hz. The sampling rate is the rate at which the optical sensors **40** receive light signals from sequentially fired LEDs **50**. In some embodiments, the light signals from the LEDs **50** are fired at different rates for each tracker **44**, **46**, **48**.

Referring to FIG. 2, each of the LEDs **50** are connected to a tracker controller **62** located in a housing (not shown) of the associated tracker **44**, **46**, **48** that transmits/receives data to/from the navigation computer **26**. In one embodiment, the tracker controllers **62** transmit data on the order of several Megabytes/second through wired connections with the navigation computer **26**. In other embodiments, a wireless connection may be used. In these embodiments, the navigation computer **26** has a transceiver (not shown) to receive the data from the tracker controller **62**.

In other embodiments, the trackers **44**, **46**, **48** may have passive markers (not shown), such as reflectors that reflect light emitted from the camera unit **36**. The reflected light is then received by the optical sensors **40**. Active and passive arrangements are well known in the art.

In some embodiments, the trackers **44**, **46**, **48** also include a gyroscope sensor **60** and accelerometer **70**, such as the trackers shown in U.S. patent application Ser. No. 14/156,856, filed on Jan. 16, 2013, entitled, "Navigation Systems And Methods For Indicating And Reducing Line-of-sight Errors", hereby incorporated by reference.

The navigation computer **26** includes a navigation processor **52**. It should be understood that the navigation processor **52** could include one or more processors to control operation of the navigation computer **26**. The processors can be any type of microprocessor or multi-processor system. The term processor is not intended to limit the scope of the invention to a single processor.

The camera unit **36** receives optical signals from the LEDs **50** of the trackers **44**, **46**, **48** and outputs to the processor **52** signals relating to the position of the LEDs **50** of the trackers **44**, **46**, **48** relative to the localizer **34**. Based on the received optical (and non-optical signals in some embodiments), navigation processor **52** generates data indi-

cating the relative positions and orientations of the trackers **44**, **46**, **48** relative to the localizer **34**.

Prior to the start of the surgical procedure, additional data are loaded into the navigation processor **52**. Based on the position and orientation of the trackers **44**, **46**, **48** and the previously loaded data, navigation processor **52** determines the position of the working end of the surgical instrument **22** and the orientation of the surgical instrument **22** relative to the tissue against which the working end is to be applied. In some embodiments, navigation processor **52** forwards these data to a manipulator controller **54**. The manipulator controller **54** can then use the data to control a robotic manipulator **56** as described in U.S. patent application Ser. No. 13/958,834, entitled, "Navigation System And Method For Removing A Volume Of Tissue From A Patient," the disclosure of which is hereby incorporated by reference.

The navigation processor **52** also generates image signals that indicate the relative position of the surgical instrument working end to the tissue. These image signals are applied to the displays **28**, **29**. Displays **28**, **29**, based on these signals, generate images that allow the surgeon and staff to view the relative position of the surgical instrument working end to the surgical site. The displays, **28**, **29**, as discussed above, may include a touch screen or other input/output device that allows entry of commands.

Referring to FIG. 3, tracking of objects is generally conducted with reference to a localizer coordinate system LCLZ. The localizer coordinate system has an origin and an orientation (a set of x-, y-, and z-axes). During the procedure one goal is to keep the localizer coordinate system LCLZ in a known position. An accelerometer (not shown) mounted to the camera unit **36** may be used to track sudden or unexpected movement of the localizer coordinate system LCLZ, as may occur when the camera unit **36** is inadvertently bumped by surgical personnel.

Each tracker **44**, **46**, **48** and object being tracked also has its own coordinate system separate from localizer coordinate system LCLZ. Components of the navigation system **20** that have their own coordinate systems are the bone trackers **44**, **46** and the instrument tracker **48**. These coordinate systems are represented as, respectively, bone tracker coordinate systems BTRK1, BTRK2, and instrument tracker coordinate system TLTR.

Navigation system **20** monitors the positions of the femur **F** and tibia **T** of the patient by monitoring the position of bone trackers **44**, **46** firmly attached to bone. Femur coordinate system is FBONE and tibia coordinate system is TBONE, which are the coordinate systems of the bones to which the bone trackers **44**, **46** are firmly attached.

Prior to the start of the procedure, pre-operative images of the femur **F** and tibia **T** are generated (or of other tissues in other embodiments). These images may be based on MRI scans, radiological scans or computed tomography (CT) scans of the patient's anatomy. These images are mapped to the femur coordinate system FBONE and tibia coordinate system TBONE using well known methods in the art. These images are fixed in the femur coordinate system FBONE and tibia coordinate system TBONE. As an alternative to taking pre-operative images, plans for treatment can be developed in the operating room (OR) from kinematic studies, bone tracing, and other methods.

During an initial phase of the procedure, the bone trackers **44**, **46** are firmly affixed to the bones of the patient. The pose (position and orientation) of coordinate systems FBONE and TBONE are mapped to coordinate systems BTRK1 and BTRK2, respectively. In one embodiment, a pointer instrument **P** (see FIGS. 1 and 2), such as disclosed in U.S. Pat.

No. 7,725,162 to Malackowski, et al., hereby incorporated by reference, having its own tracker PT (see FIG. 2), may be used to register the femur coordinate system FBONE and tibia coordinate system TBONE to the bone tracker coordinate systems BTRK1 and BTRK2, respectively. Given the fixed relationship between the bones and their bone trackers 44, 46, positions and orientations of the femur F and tibia T in the femur coordinate system FBONE and tibia coordinate system TBONE can be transformed to the bone tracker coordinate systems BTRK1 and BTRK2 so the camera unit 36 is able to track the femur F and tibia T by tracking the bone trackers 44, 46. This pose-describing data are stored in memory integral with both manipulator controller 54 and navigation processor 52.

The working end of the surgical instrument 22 (also referred to as energy applicator distal end) has its own coordinate system EAPP. The origin of the coordinate system EAPP may represent a centroid of a surgical cutting bur, for example. The pose of coordinate system EAPP is fixed to the pose of instrument tracker coordinate system TLTR before the procedure begins. Accordingly, the poses of these coordinate systems EAPP, TLTR relative to each other are determined. The pose-describing data are stored in memory integral with both manipulator controller 54 and navigation processor 52.

Referring to FIG. 2, a localization engine 100 is a software module that can be considered part of the navigation system 20. Components of the localization engine 100 run on navigation processor 52. In some versions of the invention, the localization engine 100 may run on the manipulator controller 54.

Localization engine 100 receives as inputs the optically-based signals from the camera controller 42 and, in some embodiments, the non-optically based signals from the tracker controller 62. Based on these signals, localization engine 100 determines the pose of the bone tracker coordinate systems BTRK1 and BTRK2 in the localizer coordinate system LCLZ. Based on the same signals received from the instrument tracker 48, the localization engine 100 determines the pose of the instrument tracker coordinate system TLTR in the localizer coordinate system LCLZ.

The localization engine 100 forwards the signals representative of the poses of trackers 44, 46, 48 to a coordinate transformer 102. Coordinate transformer 102 is a navigation system software module that runs on navigation processor 52. Coordinate transformer 102 references the data that defines the relationship between the pre-operative images of the patient and the bone trackers 44, 46. Coordinate transformer 102 also stores the data indicating the pose of the working end of the surgical instrument relative to the instrument tracker 48.

During the procedure, the coordinate transformer 102 receives the data indicating the relative poses of the trackers 44, 46, 48 to the localizer 34. Based on these data and the previously loaded data, the coordinate transformer 102 generates data indicating the relative position and orientation of both the coordinate system EAPP, and the bone coordinate systems, FBONE and TBONE to the localizer coordinate system LCLZ.

As a result, coordinate transformer 102 generates data indicating the position and orientation of the working end of the surgical instrument 22 relative to the tissue (e.g., bone) against which the instrument working end is applied. Image signals representative of these data are forwarded to displays 28, 29 enabling the surgeon and staff to view this information. In certain embodiments, other signals representative of

these data can be forwarded to the manipulator controller 54 to guide the manipulator 56 and corresponding movement of the surgical instrument 22.

Before using the surgical instrument 22 to treat the patient, certain preparations are necessary such as draping the patient and preparing the surgical site for treatment. For instance, in knee arthroplasty, surgical personnel may secure the leg of interest in a leg holder, and drape the patient and equipment. One such leg holder is shown in U.S. patent application Ser. No. 13/554,010, entitled, "Multi-position Limb Holder", published as U.S. Patent Application Publication No. 2013/0019883, hereby incorporated by reference.

Other preparations include placing objects needed for surgery in the operating room. Some of these objects are used in proximity to areas in which the surgical instrument 22 will maneuver. These objects can include leg holders, retractors, suction/irrigation tools, surgical personnel, and the like. During the surgery, these objects are to be avoided by the surgical instrument 22. To facilitate the avoidance of these objects during the surgery position information for one or more of these objects is determined either directly or indirectly. In some embodiments, one or more of the objects are dynamically tracked by the navigation system 20 during the surgery.

Referring to FIG. 4, in one embodiment, position information can be obtained indirectly from an object using the pointer instrument P, an example of which is disclosed in U.S. Pat. No. 7,725,162 to Malackowski, et al., hereby incorporated by reference. The pointer P has its own tracker PT with LEDs 50 that transmit signals to the camera unit 36 in the same manner as trackers 44, 46, 48. Position of a tip of the pointer P is known relative to the LEDs 50 on the pointer P and stored in the pointer P in electronic format for later transmitting to the camera unit 36 via transceivers. Alternatively, the position information for the tip is stored in the navigation computer 26 or calibrated to a known location in the field. In either case, since the tip position is known, the pointer P can be used to determine the positions of objects to be avoided by the surgical instrument 22.

Once the tip touches certain surfaces of the object, a trigger or switch (not shown) on the pointer P is actuated by the user or alternatively the tip may include a sensor that automatically senses when it is in contact with a surface. A corresponding signal is sent to the transceiver on the camera unit 36 to read the signals from the LEDs 50 on the pointer tracker PT so that the position of the tip can be calculated, which correlates to a point on the surface of the object. As more points on the surface are touched by the tip and their positions calculated by the navigation system 20, models of the object can be created to define a position and orientation of the object in the localizer coordinate system LCLZ. Such models can be created using conventional surface mapping tools and the like.

The created models are used as virtual constraint boundaries to guide movement of the surgical instrument 22. The models may be displayed on displays 28, 29 to show the locations of the objects and/or information relating to the models can be forwarded to the manipulator controller 54 to guide the manipulator 56 and corresponding movement of the surgical instrument 22 relative to these virtual constraint boundaries to prevent the object from being contacted by the surgical instrument 22.

When the object is stationary during the surgery the above method of determining position and/or orientation is suitable to provide a virtual constraint boundary, or if the object to be tracked is not stationary, but in a fixed location relative to another tracked object. However, if the object typically

moves during the surgery, additional measures are needed to enable continuous tracking of the object. In some embodiments, mountable trackers **110** may be mounted to the objects. These trackers **110** may be generic with respect to the objects and thus, not be calibrated to the objects. In this case, the trackers **110** are first attached to the objects.

One such object may be a retractor, such as the retractor assemblies **104** shown in FIG. **4**. The trackers **110** may be attached to the retractor assemblies **104** by a tracker connector located on the retractor assemblies **104**, such as those shown in U.S. Pat. No. 7,725,162 to Malackowski, et al., hereby incorporated by reference, or the trackers **110** may be mounted with conventional fasteners or clamps to fix the trackers **110** to the retractor assemblies **104**. Examples of retractor assemblies that may be used are shown in U.S. patent application Ser. No. 13/554,010, entitled, "Multi-position Limb Holder", published as U.S. Patent Application Publication No. 2013/0019883, hereby incorporated by reference. Once the tracker **110** is fixed to the retractor assembly **104**, the pointer **P** can be used to register the surfaces or other points on the retractor assembly **104**. Each tracker **110** includes three or more LEDs (not shown) that transmit signals to the camera unit **36** in the same manner as trackers **44**, **46**, **48**. The camera unit **36** and/or navigation computer **26** are then able to determine a position of each of the LEDs in the localizer coordinate system LCLZ. While the camera unit **36** is receiving signals from the LEDs on tracker **110**, the pointer **P** is used to touch on several points on the retractor assembly **104** and transmit corresponding signals to the camera unit **36** to determine position information from the pointer **P** using the pointer tracker **PT**. This enables the navigation computer **26** to associate points on the retractor assembly **104** with positions of the LEDs on the tracker **110**. Then, through a boundary creation software module (not shown) run by the navigation processor **52**, a virtual constraint boundary can be created that is associated with the retractor assembly **104** and dynamically trackable via the tracker **110**.

In some embodiments, the boundary can be created by connecting each of the captured points together. This creates a web or mesh that defines a surface boundary. If only two points are captured, the boundary may be a line between the points. If three points are captured, the boundary may be a triangle formed by lines connecting adjacent points. The displays **28**, **29** can be used to provide visual feedback of the shape of the boundary created. The input devices, e.g., mouse, touch screen, etc. could be used to modify the boundary such as by shifting the boundary, enlarging or shrinking the boundary, changing the shape of the boundary, etc. Once created, the boundary may be defined in the boundary creation software module as a virtual constraint boundary across which the surgical instrument **22** is prevented from moving in accordance with the robotic control functionality described in U.S. patent application Ser. No. 13/958,834, entitled, "Navigation System And Method For Removing A Volume Of Tissue From A Patient," the disclosure of which is hereby incorporated by reference. The manipulator controller **54** may also continuously track movement of the virtual constraint boundary and continuously adjust a path and/or orientation of the surgical instrument **22** as the virtual constraint boundary moves, to avoid the virtual constraint boundary.

The virtual constraint boundary can also be tracked simultaneously with tracking of a virtual cutting boundary associated with the femur **F** or tibia **T** described in U.S. patent application Ser. No. 13/958,834, entitled, "Navigation System And Method For Removing A Volume Of Tissue From

A Patient," the disclosure of which is hereby incorporated by reference. The virtual constraint boundary may move relative to the virtual cutting boundary during the surgery. Tracking of the boundaries would also enable tracking of the relative movement between such boundaries.

Models of the objects being tracked may be displayed on displays **28**, **29** to show the location of the objects. Representations of the virtual boundaries and the anatomy being treated may also be shown on displays **28**, **29**. Additionally, information relating to the virtual constraint boundaries and virtual cutting boundary can be forwarded to the manipulator controller **54** to guide the manipulator **56** and corresponding movement of the surgical instrument **22** relative to these virtual boundaries so that the surgical instrument **22** does not intrude on the virtual boundaries.

In some embodiments, a virtual boundary is associated with the surgical instrument **22**. The surgical instrument virtual boundary is tracked via the instrument tracker **48**. The surgical instrument virtual boundary may be defined merely by a model of the surgical instrument **22**. The manipulator controller **54** then monitors movement of the surgical instrument virtual boundary relative to the other virtual constraint boundaries, including the virtual cutting boundaries and other virtual constraint boundaries associated with other objects. The manipulator controller **54** is then programmed to continuously track movement of the boundaries and update guidance of the surgical instrument **22** as the boundaries move relative to the surgical instrument **22**.

Objects to be avoided by the surgical instrument **22** in the operating room may be tracked indirectly by associating the object with one or more trackers that are not directly fixed to the object. For instance, in FIG. **4**, the opening **106** in the tissue, although not directly attached to a tracker, is formed by the retractor assemblies **104** with trackers **110** fixed thereto. Since the retractor assemblies **104** form the opening **106**, there is a general correlation between the size and shape of the opening **106** and the position and orientation of the retractor assemblies **104**, which can be tracked by the navigation system **20** using the trackers **110**, as described above. Therefore, the opening **106** can also be dynamically tracked.

The opening **106** can be defined in the boundary creation software module using the points associated with the retractor assemblies **104** since the opening **106** lies along an edge of the retractor assemblies **104**. Alternatively, the opening **106** can be traced using the pointer **P**. In the latter case, the pointer **P** is used to capture points defining a periphery of the opening **106** such that the points can be connected in the boundary creation software module to form a ring representing the opening **106**. The ring may be defined in the boundary creation software module as a virtual constraint boundary to constrain movement of the surgical instrument **22** to within the ring in accordance with the robotic control functionality associated with such openings described in U.S. patent application Ser. No. 13/958,834, entitled, "Navigation System And Method For Removing A Volume Of Tissue From A Patient," the disclosure of which is hereby incorporated by reference. The opening **106** could additionally be registered to the trackers **110** so that movement of the opening **106** is trackable using the trackers **110**. Other tissues to be avoided by the surgical instrument **22** such as nerve tissue, ligaments, and the like can similarly be outlined by the pointer **P** and associated with the trackers **110** to track their movement.

Referring to FIG. **5**, a leg holder **200** for supporting a leg of a patient is shown. The leg holder **200** is described in

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more detail in U.S. patent application Ser. No. 13/554,010, entitled, "Multi-position Limb Holder", published as U.S. Patent Application Publication No. 2013/0019883, hereby incorporated by reference. An alternative retractor assembly **105** for attaching to the leg holder **200** is shown in FIG. 5. The alternative retractor assembly **105** is described in more detail in U.S. patent application Ser. No. 13/554,010, hereby incorporated by reference.

Retractor heads **107**, **109** in FIGS. 6 and 7 can be used to retract soft tissue to access bone in a surgical procedure. Use of these types of heads **107**, **109** for retracting tissue is described in more detail in U.S. patent application Ser. No. 13/554,010, hereby incorporated by reference. In FIGS. 6 and 7, tracking elements are fixed to the heads **107**, **109** so that the heads **107**, **109** can be tracked by the navigation system **20**. In the embodiment shown, the tracking elements are three or more LEDs **50** that are integrated into the structure of each of the heads **107**, **109** and fixed in relationship to one another. The geometric model of each head **107**, **109** in relation to the LEDs **50** is also stored on the retractor head **107**, **109** in memory (not shown) and can be transmitted to the camera unit **36** via transceivers (including transceiver, not shown, integrated into the retractor head **107**, **109**). Alternatively, the model of each head is pre-stored in the navigation computer **26** and accessed during navigation setup by identifying a type or serial no. of the retractor head **107**, **109** using the boundary creation software module. The shape of each retractor head **107**, **109** can also be identified by correlating a unique LED pattern on the retractor head **107**, **109** to a database of retractor head shapes.

By creating virtual constraint boundaries associated with the shapes of the retractor assemblies **104** and tracking movement of the virtual constraint boundaries using trackers **110** or integrated tracking elements, the manipulator controller **54** can guide movement of the surgical instrument **22** with respect to the retractor virtual constraint boundaries and the virtual cutting boundaries so that the surgical instrument **22** is not moved beyond these boundaries thereby avoiding inadvertent contact with the retractor assemblies **104** or with bone or other tissue to remain after the surgery. These virtual boundaries may be used in both a manual mode and semi-autonomous mode of the surgical manipulator as described in U.S. patent application Ser. No. 13/958,834, entitled, "Navigation System And Method For Removing A Volume Of Tissue From A Patient," the disclosure of which is hereby incorporated by reference.

Referring to FIG. 8, a flexible shape sensing device **300** may also be used to determine a position of an object, such as opening **106**. The flexible shape sensing device **300** includes a housing **302** having its own shape sensing coordinate system SS. The housing **302** forms part of a reflectometer, such as a Luna Distributed Sensing System commercially available from Luna Innovations Incorporated of Roanoke, Virginia. Another example of a commercially available reflectometer is the Optical Backscatter Reflectometer from Luna Innovations Incorporated.

A fiber optic cable **304** extends from the housing **302** and is laid on the patient's skin about the opening **106** in close proximity to the opening **106**. In some embodiments, the cable **304** is adhered to the skin in a perimeter with an offset from the opening **106**. In some embodiments, the offset is less than five millimeters from the opening **106** at all locations along the perimeter of the opening **106**. In other embodiments, different offsets may be used or the offsets may be measured after placing the fiber optic cable **304** so that the location of the fiber optic cable **304** relative to the

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opening **106** is known. The cable **304** is flexible so that as the shape of the opening **106** changes, the shape of the cable **304** also changes. Position of the cable **304** is able to be dynamically tracked. The flexible shape sensing device **300** including the reflectometer, cable, and other features, and their method of use for determining position are described in U.S. Pat. No. 7,772,541 to Froggatt et al., hereby incorporated by reference.

Tracking elements, such as LEDs **50** may be integrated into the flexible shape sensing device **300**. Alternatively, a tracker (not shown) can be mounted to the housing **302**. The LEDs **50** integrated into the flexible shape sensing device **300** transmit signals to the camera unit **36** in the same manner as the LEDs **50** of the trackers **44**, **46**, **48**. Accordingly, the position and orientation of the housing **302** and the shape sensing coordinate system SS can be determining by the navigation system **20** in the localizer coordinate system LCLZ. Movement of the cable **304** results in changes in position in shape sensing coordinate system SS, which is fixed with respect to housing **302**. Coordinate system SS is registered to the localizer coordinate system LCLZ using the LEDs **50** on the housing **302**. Once registered, changes in position of the cable **304** can also be determined in the localizer coordinate system LCLZ.

The opening **106** may be defined in the boundary creation software module as a virtual constraint boundary to constrain movement of the surgical instrument **22** to within the opening **106** in accordance with the robotic control functionality associated with such openings described in U.S. patent application Ser. No. 13/958,834, entitled, "Navigation System And Method For Removing A Volume Of Tissue From A Patient," the disclosure of which is hereby incorporated by reference. Other tissues to be avoided by the surgical instrument **22** such as nerve tissue, ligaments, and the like can similarly be tracked using flexible shape sensing devices **300**. Likewise, flexible shape sensing devices **300** could be used to establish other boundaries, such as being integrated into gloves worn by the surgical staff so that boundaries associated with surgical personnel can be created.

Machine vision can identify objects in the operating room and create virtual constraint boundaries associated with the objects. FIG. 9 shows a machine vision system **400**. Machine vision system **400** includes a 3-dimensional machine vision camera **402**. The vision camera **402** is arranged so that a field-of-view of the vision camera **402** encompasses the surgical site and objects in proximity to the surgical site. As shown in FIG. 9, such objects may include the surgical instrument **22** (shown as a cutting bur), retractor assemblies **104**, femur F, and tibia T. The machine vision system **400** has a control unit (not shown) in communication with the vision camera **402**. The control unit includes a processor, memory, and storage and is in communication with the navigation computer **26**.

Initially, the objects to be tracked are identified. The objects may be identified by selecting objects stored in memory on the control unit using machine vision software. For instance, groups of pixels associated with different sizes and shapes of retractor assemblies **104** may be stored in the control unit. By selecting one of the retractor assemblies **104** to be tracked the machine vision software identifies the corresponding group of pixels and the machine vision software then operates to detect like groups of pixels using conventional pattern recognition technology.

Alternatively, the objects can be identified using an interface in which a user outlines or selects the objects to be tracked on the displays **28**, **29**. For instance, images taken by

the vision camera 402 from overhead the surgical site—similar to the image shown in FIG. 9—are displayed on the displays 28, 29. The user then, using a mouse, digital pen, or the like, traces objects to be tracked on the display 28, 29. The machine vision software stores the pixels associated with the object that was traced into its memory. The user identifies each object by a unique identifier such as naming the object “MEDIAL RETRACTOR”, etc. in the machine vision software so that the saved group of pixels is now associated with the unique identifier. Multiple objects could be stored in this manner. The machine vision system 400 utilizes conventional pattern recognition and associated software to later detect these objects.

The machine vision system 400 is able to detect movement of these objects by continuously taking images, reviewing the images, and detecting movement of the groups of pixels associated with the objects. In some cases, position information from the control unit of the machine vision system 400 for the objects can be transmitted to the navigation computer 26. Likewise, position information from the navigation computer 26 can be transmitted from the navigation computer 26 to the control unit of the machine vision system 400.

Control unit of the machine vision system 400 may provide position information for the objects in a machine vision coordinate system MV. The vision camera 402 also includes LEDs 50 so that the camera unit 36 can track and thus register the position and orientation of the machine vision coordinate system MV relative to the localizer coordinate system LCLZ. Thus, position information from the vision camera 402 can be determined in the localizer coordinate system LCLZ. Virtual boundaries can thus be associated with the objects in the machine vision system 400 and information relating to these virtual boundaries can be communicated to the navigation computer 26. Additionally, information relating to the virtual constraint boundaries can be forwarded to the manipulator controller 54 to guide the manipulator 56 and corresponding movement of the surgical instrument 22 relative to these virtual boundaries.

The objects can also be initially registered to the localizer coordinate system LCLZ using the pointer P. For instance, when the retractor assemblies 104 are not equipped with trackers 110 or integrated tracking elements, the pointer P may be used to initially establish virtual constraint boundaries associated with the retractor assemblies 104 when the retractor assemblies 104 are at rest, i.e., not moving. These virtual constraint boundaries would then be stored in the navigation computer 26 and/or manipulator controller 54 for use in guiding the robotic manipulator 56. The machine vision system 400 would also be configured to detect movement of the retractor assemblies 104 as previously described, i.e., by tracking movement of the groups of pixels associated with the retractor assemblies 104.

Machine vision detection of movement of a retractor assembly 104 could then be used to shift the virtual constraint boundary stored in the navigation computer for the retractor assembly 104 by defining a change in pose of the retractor assembly 104 (e.g., translation along 3 axes/rotation about 3 axes). The machine vision system 400 would operate to establish a first pose of the retractor assembly 140 at time t1 and a second pose at time t2. The difference in pose between t1 and t2 would be provided to the navigation computer 26 and/or manipulator controller 54 to move the associated virtual constraint boundary by a proportional amount in the localizer coordinate system LCLZ. In some

computer 26 and/or manipulator controller 54 to update a position of the retractor assembly 104.

In some embodiments, the robotic system is a robotic surgical cutting system for cutting away material from a patient's anatomy, such as bone or soft tissue. Once the cutting system is determined to be in the proper position by the navigation system 20, the cutting system cuts away material to be replaced by surgical implants such as hip and knee implants, including unicompartmental, bicompartamental, or total knee implants. Some of these types of implants are shown in U.S. patent application Ser. No. 13/530,927, entitled, “Prosthetic Implant and Method of Implantation”, the disclosure of which is hereby incorporated by reference. The navigation system 20 instructs the surgeon on proper procedures for locating these implants on bone and securing the implants in position, including the use of trial implants.

In other systems, the instrument 22 has a cutting tool that is movable in three degrees of freedom relative to a handheld housing and is manually positioned by the hand of the surgeon, without the aid of cutting jigs, guide arms or other constraining mechanism. Such systems are shown in U.S. patent application Ser. No. 13/600,888, entitled, “Surgical Instrument Including Housing, a Cutting Accessory that Extends from the Housing and Actuators that Establish the Position of the Cutting Accessory Relative to the Housing”, the disclosure of which is hereby incorporated by reference.

In these embodiments, the system includes a hand held surgical cutting instrument having a cutting tool. A control system controls movement of the cutting tool in at least three degrees of freedom using internal actuators/motors, as shown in U.S. patent application Ser. No. 13/600,888, entitled, “Surgical Instrument Including Housing, a Cutting Accessory that Extends from the Housing and Actuators that Establish the Position of the Cutting Accessory Relative to the Housing”, the disclosure of which is hereby incorporated by reference. The navigation system 20 communicates with the control system. One tracker (such as tracker 48) is mounted to the instrument. Other trackers (such as trackers 44, 46) are mounted to a patient's anatomy. The navigation system 20 communicates with the control system of the hand held surgical cutting instrument. The navigation system 20 communicates position and/or orientation data to the control system. The position and/or orientation data is indicative of a position and/or orientation of the instrument 22 relative to the anatomy. This communication provides closed loop control to control cutting of the anatomy such that the cutting occurs within a predefined boundary (the term predefined boundary is understood to include predefined trajectory, volume, line, other shapes or geometric forms, and the like).

In some embodiments, a 3-D video camera (not shown) is attached to the camera unit 36. The video camera is oriented such that a field of view of the camera unit 36 can be associated with the field of view of the video camera. In other words, the two fields of view may be matched or otherwise correlated such that if an object can be seen in video images streamed from the video camera, the objects are also within the field of view of the camera unit 36. A coordinate system of the video camera can also be transformed into the localizer coordinate system LCLZ or vice versa so that positions and/or orientations of objects shown in the video images streamed from the video camera are known in the localizer coordinate system LCLZ. Video images from the video camera can be streamed to the displays 28, 29 and the user can then identify on the displays 28, 29, using an input device, such as a mouse or touch screen, virtual constraint boundaries to delineate zones to be

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avoided by the instrument 22. The video images could be provided in 2-D or in 3-D to facilitate the creation of these virtual constraint boundaries. Information relating to the positions and/or orientation of these virtual constraint boundaries would be provided into the localizer coordinate system LCLZ and tracked by the navigation computer 26 or manipulator controller 54, for example, to prevent the instrument 22 from intruding on the boundaries created.

In some embodiments, when the manipulator controller 54 or navigation computer 26 detect that the instrument 22 is approaching one of the virtual constraint boundaries, an alarm may be generated. The alarm may include visual, tactile, or audible feedback to the user that indicates to the user that the object associated with the virtual constraint boundary is about to be struck and/or may include visual, tactile, or audible indications of distance from the object or associated virtual constraint boundaries.

Several embodiments have been discussed in the foregoing description. However, the embodiments discussed herein are not intended to be exhaustive or limit the invention to any particular form. The terminology which has been used is intended to be in the nature of words of description rather than of limitation. Many modifications and variations are possible in light of the above teachings and the invention may be practiced otherwise than as specifically described.

What is claimed is:

1. A surgical system comprising:
 - a robotic manipulator configured to support and move a cutting instrument;
 - a navigation system including a localizer, a first tracker coupled to the robotic manipulator, and a second tracker coupled to a bone at a surgical site, wherein the navigation system is configured to detect the first tracker to monitor a pose of the cutting instrument and the second tracker to monitor a pose of the bone;
 - a machine vision system including a vision camera; and
 - a control system coupled to the robotic manipulator, the localizer, and the machine vision system, wherein the control system is configured to:
 - control the robotic manipulator to move the cutting instrument along a path to cut away material from the bone at the surgical site based on the monitored pose of the cutting instrument and the monitored pose of the bone;
 - detect, with the machine vision system, an object at, or in proximity to, the surgical site;
 - associate a virtual boundary with the object detected by the machine vision system;
 - adjust the path of the cutting instrument based on the virtual boundary; and
 - control the robotic manipulator to move the cutting instrument along the adjusted path to enable the cutting instrument to avoid the object.
2. The surgical system of claim 1, wherein, to enable the cutting instrument to avoid the object, the control system is further configured to:
 - control the robotic manipulator to adjust an orientation of the cutting instrument based on the virtual boundary.
3. The surgical system of claim 1, wherein the control system is further configured to track movement of the virtual boundary.
4. The surgical system of claim 3, wherein the control system adjusts the path of the cutting instrument further in response to tracked movement of the virtual boundary.

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5. The surgical system of claim 3, wherein the control system is further configured to track movement of the virtual boundary with the vision camera of the machine vision system.

6. The surgical system of claim 5, wherein the control system tracks movement of the virtual boundary with the vision camera by further being configured to:

- receive images acquired by the vision camera;
- identify groups of pixels in the images that are associated with the object; and
- detect movement of the groups of pixels in the images.

7. The surgical system of claim 1, wherein the object is a retractor configured to retract tissue adjacent to the bone.

8. The surgical system of claim 7, wherein the control system associates the virtual boundary with the retractor by further being configured to:

- identify the retractor with the machine vision system;
- retrieve, from non-transitory memory, a geometric model of the retractor; and
- utilize the geometric model to define the virtual boundary.

9. The surgical system of claim 7, wherein the control system detects the retractor with the machine vision system by being configured to detect a tracker or tracking element coupled to the retractor.

10. The surgical system of claim 1, wherein the object is a first object and the virtual boundary is a first virtual boundary associated with the first object, and wherein the control system is further configured to:

- detect, with the machine vision system, a second object at, or in proximity to, the surgical site; and
- associate a second virtual boundary with the second object.

11. The surgical system of claim 10, wherein the control system is further configured to:

- track movement of the first virtual boundary;
- track movement of the second virtual boundary; and
- track relative movement between the first virtual boundary and the second virtual boundary.

12. The surgical system of claim 10, wherein:

- the first object is a surgical instrument to be avoided by the cutting instrument or a soft tissue structure to be avoided by the cutting instrument; and
- the second object is the bone.

13. The surgical system of claim 12, wherein:

- the second virtual boundary is a virtual constraint boundary delineating a first region of the bone to be treated by the cutting instrument from a second region of the bone to be avoided by the cutting instrument; and
- the control system is further configured to control the robotic manipulator to avoid interaction of the cutting instrument with the virtual constraint boundary.

14. The surgical system of claim 12, wherein the surgical instrument is a retractor configured to retract tissue adjacent to the bone.

15. The surgical system of claim 1, wherein the navigation system is coupled to the control system and comprises a display device, and wherein the navigation system is further configured to:

- receive a pose of the object from the machine vision system;
- present, on the display device, graphical representations of: the cutting instrument, the bone, and the virtual boundary; and
- update a relative spatial relationship of the graphical representations based on the poses of the cutting instrument, the bone, and the object.

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16. The surgical system of claim 1, wherein the control system controls the robotic manipulator to move the cutting instrument along the path by further being configured to operate the robotic manipulator in a semi-autonomous mode.

17. The surgical system of claim 1, wherein the cutting instrument is a rotating cutting bur.

18. The surgical system of claim 1, wherein the bone is a femur bone or a tibia bone, and wherein the control system controls the robotic manipulator to move the cutting instrument along the path to cut away material to prepare the femur bone or the tibia bone to receive a surgical implant for a joint arthroplasty procedure.

19. A method of operating a surgical system, the surgical system including a robotic manipulator configured to support and move a cutting instrument, a navigation system including a localizer, a first tracker coupled to the robotic manipulator, and a second tracker coupled to a bone at a surgical site, a machine vision system including a vision camera, and a control system coupled to the robotic manipulator, the localizer, and the machine vision system, the method comprising the control system:

monitoring a pose of the cutting instrument based on detection of the first tracker by the localizer and a pose of the bone based on detection of the second tracker by the localizer;

controlling the robotic manipulator for moving the cutting instrument along a path for cutting away material from the bone at the surgical site based on the monitored pose of the cutting instrument and the monitored pose of the bone;

detecting, with the machine vision system, an object at, or in proximity to, the surgical site;

associating a virtual boundary with the object detected by the machine vision system;

adjusting the path of the cutting instrument based on the virtual boundary; and

controlling the robotic manipulator for moving the cutting instrument along the adjusted path and for enabling the cutting instrument to avoid the object.

20. A surgical system comprising:

a robotic manipulator configured to support and move a cutting instrument;

a navigation system including a localizer, a first tracker coupled to the robotic manipulator, and a second tracker coupled to a bone at a surgical site, wherein the

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navigation system is configured to detect the first tracker to monitor a pose of the cutting instrument and the second tracker to monitor a pose of the bone;

a machine vision system including a vision camera; and a control system coupled to the robotic manipulator, the localizer, and the machine vision system, wherein the control system is configured to:

control the robotic manipulator to move the cutting instrument to cut away material from the bone at the surgical site based on the monitored pose of the cutting instrument and the monitored pose of the bone;

detect, with the machine vision system, an object at, or in proximity to, the surgical site;

associate a virtual boundary with the object detected by the machine vision system; and

control the robotic manipulator to automatically adjust an orientation of the cutting instrument based on the virtual boundary to enable the cutting instrument to avoid the object.

21. A method of operating a surgical system, the surgical system including a robotic manipulator configured to support and move a cutting instrument, a navigation system including a localizer, a first tracker coupled to the robotic manipulator, and a second tracker coupled to a bone at a surgical site, a machine vision system including a vision camera, and a control system coupled to the robotic manipulator, the localizer, and the machine vision system, the method comprising the control system:

monitoring a pose of the cutting instrument based on detection of the first tracker by the localizer and a pose of the bone based on detection of the second tracker by the localizer;

controlling the robotic manipulator for moving the cutting instrument for cutting away material from the bone at the surgical site based on the monitored pose of the cutting instrument and the monitored pose of the bone;

detecting, with the machine vision system, an object at, or in proximity to, the surgical site;

associating a virtual boundary with the object detected by the machine vision system; and

controlling the robotic manipulator for automatically adjusting an orientation of the cutting instrument based on the virtual boundary to enable the cutting instrument to avoid the object.

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